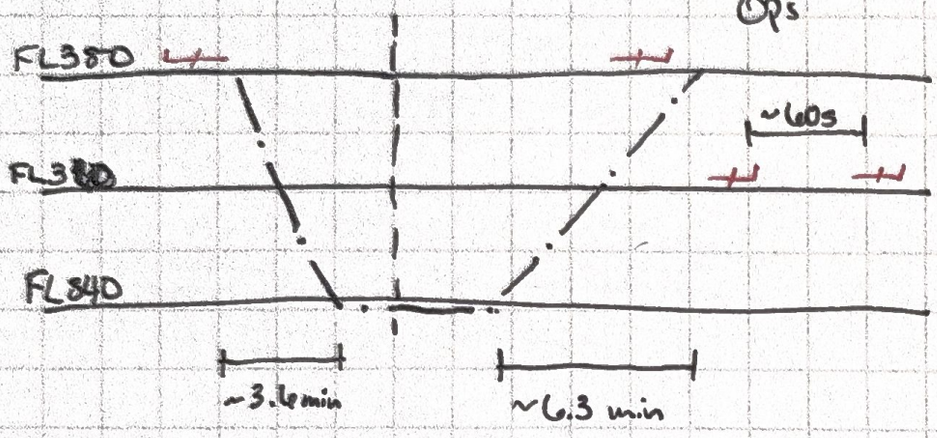
Check "architecture model" "decision-by-decision" if decider has needed info.

- Diagram w/ports
- So, objective → requirement → Activity → Sequence → exchanges w/info

Nominal Ops | Conflict Resolution | Controller w/o GW, ACM



Decision 12 min before. Assume decision + radio occurs < 1 min
- Descend 11 → 7.4 min remaining.
- Level ~10 min to ensure clearance w/ AC3 & AC4.
- Climb 6.3 min to cover FL340 → FL380
- Level for 5 min?

$$C1 = \frac{\text{lbs fuel}}{\text{minute}} \quad \frac{\$/\text{hr}}{\$/100\text{lbs}} \quad \frac{100\ \text{lbs}}{1\ \text{hr}}$$

Captain + FO + Cabin Crew + Maintenance Cost + Schedule Pressure
400k/yr 250 250 ? ?

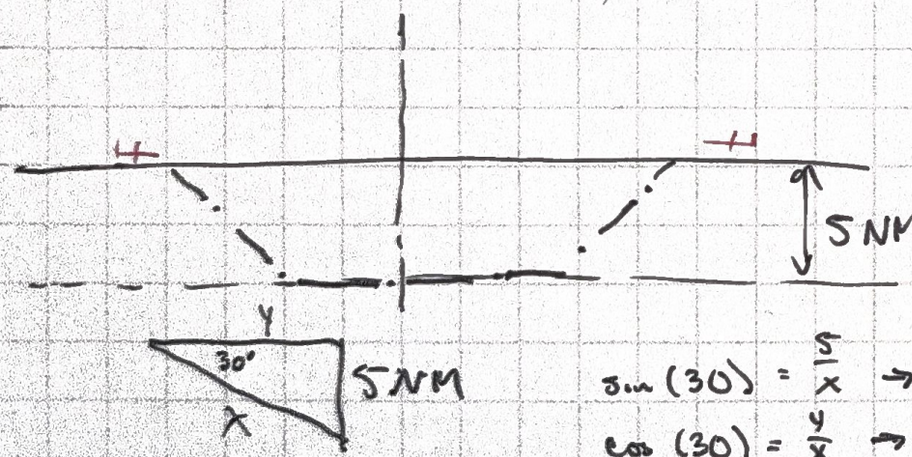
~$1M/yr for 5 people each @ ~2080 hrs/yr
*2 for underestimate of cabin crew salary + ?
→ ~$3/min

Assume $2.50 per kg @ airports for Jet-A

I've heard typical is ~30 * $\frac{100\ \text{lbs}}{\text{hr}}$ = $\frac{50\ \text{lbs}}{\text{min}}$
= 22.67 kg/min
= ~57 USD/min

Assume 15 C1 → 1520 lbs/hr

Assume AC3 & 4 very close behind AC1, so require ~14 min to clear point of conflict.



$\sin(30) = \frac{5}{x} \rightarrow x = \frac{5}{0.5} = 10$

$\cos(30) = \frac{y}{x} \rightarrow y = 8.66$

Dist $\Delta = 2*(x-y)$

So, minimum distance add w/ 30° is 2.68 NM.

If AC3 is 2 NM "behind" AC1, then 3+ NM must be made up by speed. @ 91 NM in 12 min. If 3+ NM are required, then 89 NM will be traversed in 12 min which is ~440 KIAS, which is 0.766 → 0.04 Δ.

GE Aerospace